

Assessment of Radiologically Tissue-Equivalent Materials for Breast Imaging Phantoms Using X-ray Spectrometry

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INTRODUCTION

Castable and thermoplastic polymeric materials were evaluated as tissue-equivalent materials (TEMs) for breast phantom fabrication in the diagnostic energy range (15 - 50 keV). According to the ICRU, the linear attenuation coefficient $\mu(E)$ of a TEM should not deviate by more than 5% from that of the corresponding reference tissue within the energy range¹.

Purpose

To determine attenuation properties of substitute materials for use in breast imaging phantoms through X-ray spectrometry

MATERIALS AND METHODS

Formulations for TEMs were obtained using the TEMPy toolkit². The $\mu(E)$, were experimentally determined according to the exponential attenuation law. The $\mu(E)$ of the reference tissues were obtained using the XCOM database (Figures 1 and 2).

$$\mu(E) = -\frac{1}{x} \ln \left[\frac{I(E) - I_{bg}(E)}{I_0(E) - I_{bg}(E)} \right] + \mu_{ar}$$

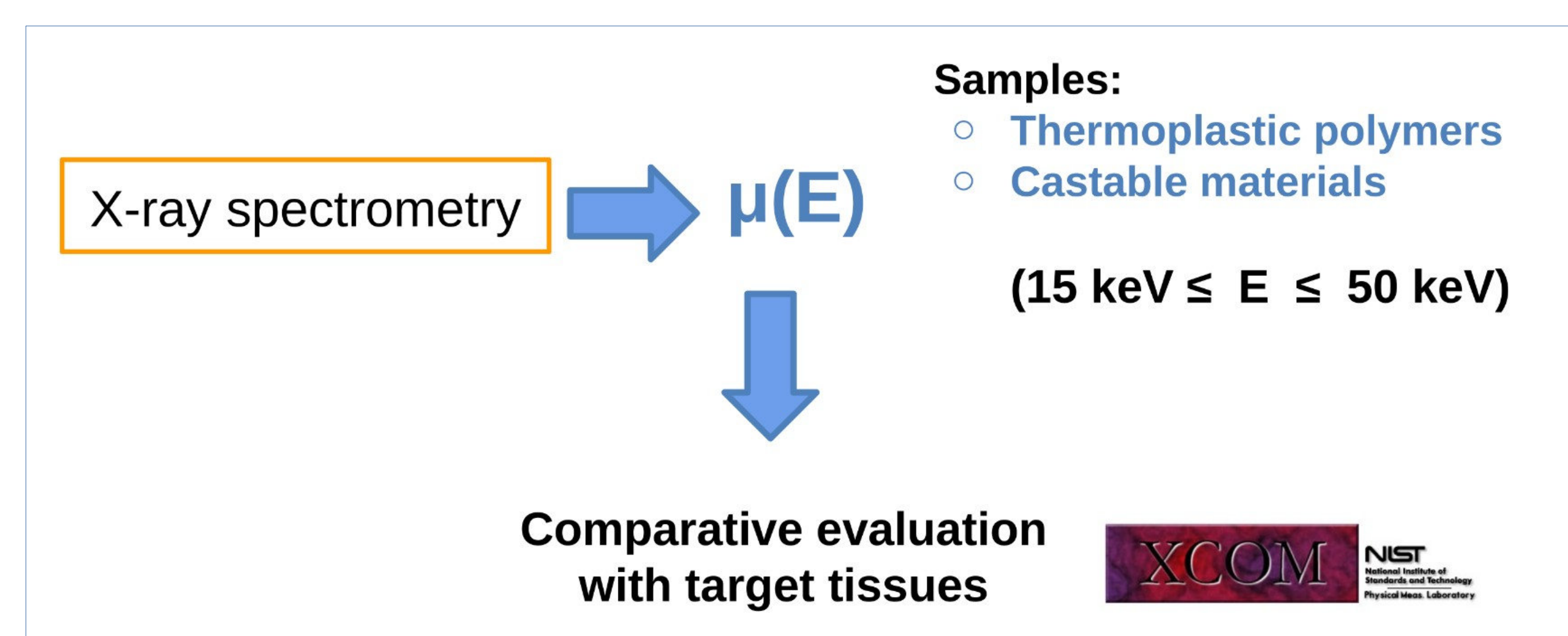


Figure 1: Sample analysis process. The linear attenuation coefficient, $\mu(E)$, of the reference tissues were obtained using the XCOM database.

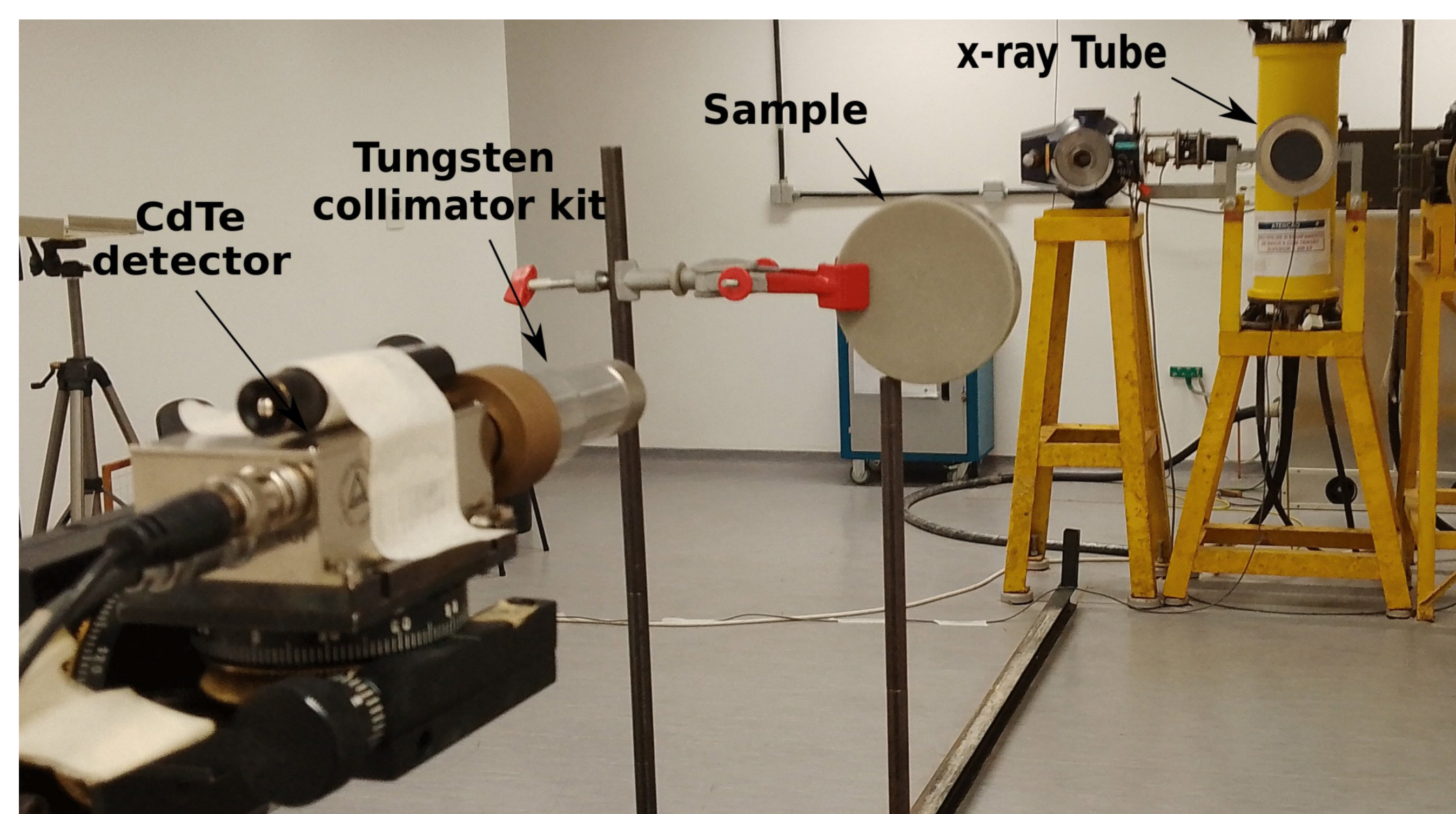


Figure 2: Experimental setup used to measure primary and transmitted x-ray spectra. We used a Philips MCN 421 x-ray tube, 3x3x1 mm³ Cadmium Telluride (CdTe) spectrometer (model XR-100T, Amptek, Inc, EUA) with the PX4 digital pulse processor³.

RESULTS

- Materials analyzed and mean percentage difference of the measured data against reference tissue in relevant energy range, Table 1.
- Linear attenuation coefficient of the reference material and measured sample, Figure 3.

Table 1: Materials analyzed as substitutes for adipose and glandular/skin tissue. For glandular tissue, mixtures of epoxy resins (ER) with additives were studied. The weight percentages of resin and additives are presented. Relative percentage difference (Diff %) of the measured data against reference tissue in relevant energy range are also shown.

Adipose		Glandular/Skin	
Material	Diff %	Material	Diff %
Epoxy resin	15	98%ER + 2%TiO ₂	3
Paraffin	33	98%ER + 2%MgO	4
Coconut wax	30	99%ER + 1%CaCO ₃	9
Polypropylene	38	99%ER + 1%NaCl	7

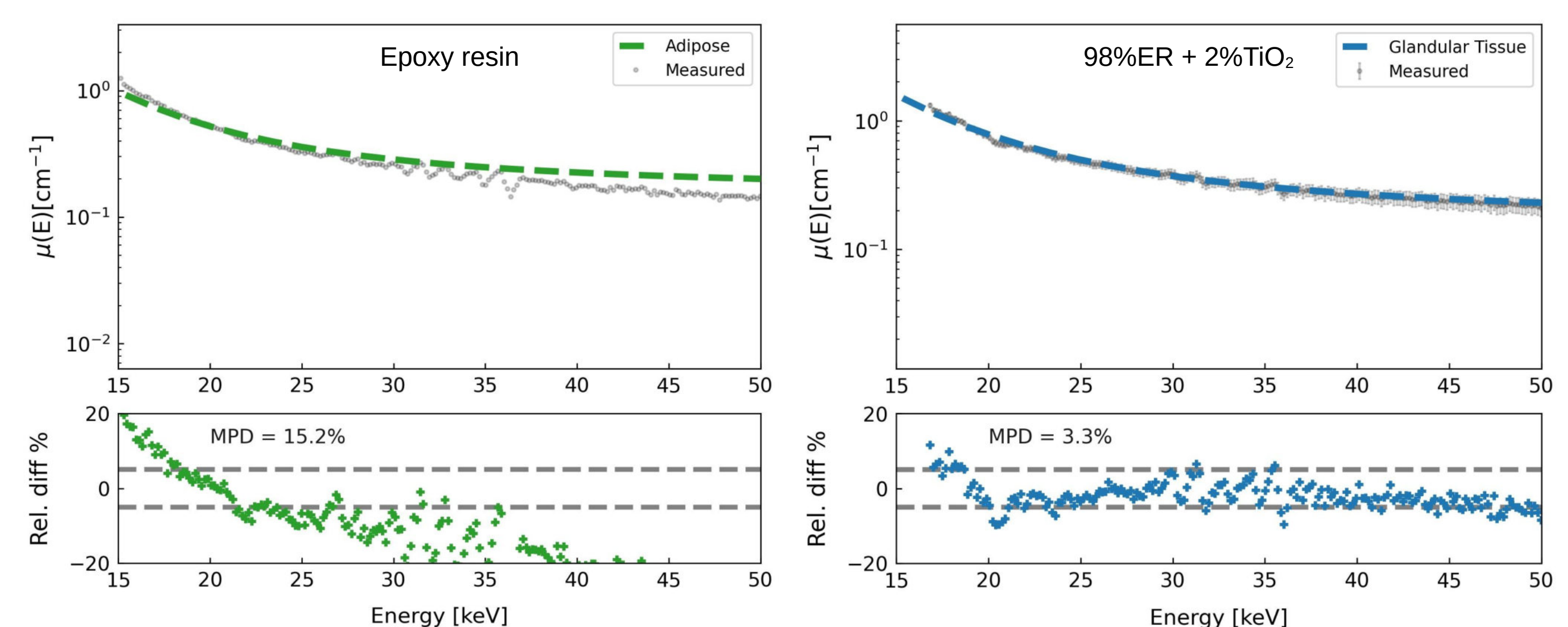


Figure 3: Linear attenuation coefficient of the reference material and measured samples. Relative percentage difference of the measured data against reference tissue are also shown with the Mean Percentage Difference (MPD). The dashed horizontal lines correspond to $\pm 5\%$.

CONCLUSIONS

The experimental methodology enabled the evaluation of the suitability of the studied materials as TEMs. The results demonstrate the feasibility of manufacturing physical breast phantoms using accessible and low-cost materials.

ACKNOWLEDGEMENTS

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